ChatGPT

ARC Prize 2025 & Human Fluid Intelligence: research and blueprint

1. 2025 ARC Prize & ARC-AGI-2: rules, tasks and scores

Competition structure. ARC Prize 2025 is hosted on Kaggle and runs from **26 Mar 2025** to **3 Nov 2025** with winners announced on **5 Dec 2025** [1]. The contest uses the **ARC-AGI-2** dataset, a harder successor to ARC-AGI-1, and aims to inspire systems capable of **fluid intelligence** rather than rote memorization. Contestants must upload code before seeing the private evaluation score and abide by new open-source requirements [2]. The grand prize is unlocked when the first eligible open-source system reaches **≥85 % accuracy on the private evaluation** [3].

Task format. Each ARC task is a small set of `train` **input/output pairs** and one or more `test` inputs contained in JSON files. The solver must infer the rule from the train examples and output **pixel-perfect predictions** for every test input. ARC-AGI-2 uses a `pass@2` scoring scheme—each test input requires exactly **two candidate solutions**. A solver gets credit for that output if either attempt is correct; this matches the study showing that **every ARC-AGI-2 task was solved by at least two humans in ≤2 attempts** [4]. The competition therefore mandates that submissions include two predictions per test input. Evaluation sets are **calibrated**: 120 public tasks, 120 semi-private tasks (used for leaderboard during the contest) and 120 private tasks (used for final scoring) [5]. Controlled human testing ensures that all evaluation tasks are IDD (independent and identically distributed) and solvable by at least two humans in pass@2 [6].

Compute and submission logistics. Kaggle enforces offline execution with a compute cost limit (~$0.42 per task) [2]. Participants must deliver a single `submission.json` containing exactly two outputs (`attempt_1` and `attempt_2`) for each test input. Code must run without internet access and must be published under an open-source license before receiving the final private evaluation score [7].

Benchmark scores. The 2024 ARC Prize showed large gains: the state-of-the-art score on ARC-AGI-1 rose from **33 % to 55.5 %** thanks to **neural-guided program synthesis** and **test-time training (TTT)** [8]. The winning team **ARChitects** scored **53.5 %** and open-sourced their approach; **MindsAI** attained **55.5 %** but did not open-source and thus was ineligible [9]. However, those systems over-fit to ARC-AGI-1: early reports indicate that their scores drop to low single digits on ARC-AGI-2. A research-review article states that **even frontier systems such as OpenAI's "o3" drop from 75.7 % on ARC-AGI-1 to 4 % on ARC-AGI-2** [10]. The ARC Prize 2025 thus calls for fundamentally better reasoning rather than scaling existing methods.

2. Human fluid intelligence and novel reasoning

Fluid intelligence is the ability to solve new problems without relying on learned facts. Cognitive neuroscience attributes fluid reasoning to a **fronto-parietal "multiple-demand" (MD) network** and several supporting systems:

1. **Multiple-demand (MD) network.** A bilateral network covering **lateral and dorsomedial frontal regions, anterior insula and intraparietal sulcus** forms an integrated system [11]. Selective damage to these regions causes **fluid-intelligence deficits** [12]. A large fMRI study showed that **individual differences in executive abilities and fluid intelligence are *positively* associated with the level of MD activity**, even after controlling for language-network activity [13]. This resolves previous conflicting findings and confirms that higher MD activation during challenging tasks predicts better fluid reasoning.

2. **Basal ganglia & working-memory gating.** Cortico-basal-ganglia circuits act as **gates on working memory**: they decide what information enters, what leaves and when to shift representations. Evidence shows that the basal ganglia serve as a gate for selecting **motor and cognitive representations** and can re-allocate memory representations [14]. This gating mechanism underlies flexible rule selection and chunking in problem solving.

3. **Parietal cortex.** Inferior parietal regions represent **relational structure**, numerosity and spatial relations [11]. They support object grouping and symmetry detection—key priors in ARC tasks.

4. **Hippocampus–medial prefrontal cortex (mPFC) loop.** The hippocampus rapidly binds new relations, while the mPFC integrates new experiences into existing schemas. During learning, the mPFC is active from the outset; its damage blocks the development and retrieval of schemas [15]. The mPFC also accommodates new information into existing schemas and resolves conflicts between old and new knowledge [16]. This interplay enables **meta-learning**—the rapid acquisition of new rules from a few examples.

5. **Insight and oscillatory dynamics.** Insightful problem solving involves a burst of high-frequency **gamma activity** in the right temporal cortex at the moment a solution becomes consciously available [17]. This is often preceded by an **alpha-band burst** that suppresses irrelevant information [18]. Such oscillations reflect gating and integration processes that precede the "aha!" moment.

Steps when a human solves a novel ARC-like puzzle

1. **Scene parsing & segmentation:** Visual cortex and parietal areas segment the grid into discrete objects and extract low-level features (colors, shapes, adjacency). Core knowledge priors—objectness, symmetry, repetition and numerosity—are invoked.
2. **Hypothesis generation:** The MD network composes candidate transformations (e.g., rotate, recolor, replicate) based on detected priors. The hippocampus retrieves similar past experiences to bias the search.
3. **Working-memory gating & evaluation:** Basal-ganglia circuits gate which transformation to test; the MD network simulates and evaluates candidate rules against the demonstrations.
4. **Meta-learning & adaptation:** The mPFC–hippocampal loop rapidly adapts internal weights or schemas to fit the few examples, chunking new rules into working memory.

5. **Insight & decision:** Synchronized high-frequency activity signals convergence on a compact rule; parietal regions apply it to the novel test input, producing a solution.

These steps highlight the interplay of perception, rule composition, gating, adaptation and insight—elements we should emulate in our solver.

3. From neuroscience to solver design

The solver should mirror the brain's modular processes:

1. **Object-centric parser (scene graph).** Emulating parietal segmentation, we build connected-component representations for each grid: bounding boxes, color masks and histograms. This provides a **relational scaffold** for reasoning about symmetries, repetition and object movement.

2. **Compact DSL of primitives.** Like human priors, the solver uses a small set of composable operations: rotations, flips, transpositions, translations, padding, cropping and color mapping. Compositionality allows building complex transformations from simple rules.

3. **Neural-guided program search (MD analog).** A discrete search over DSL sequences is guided by heuristics. A small classifier (trained on the 1 000 public training tasks) predicts which primitive families are likely relevant; these predictions gate which programs enter the beam (basal-ganglia analog). This is analogous to the MD network prioritizing candidate rules.

4. **Test-time adaptation (mPFC–hippocampus analog).** For each new task, a lightweight head is fine-tuned on the demonstration pairs to reweight heuristic scores. This adaptation models rapid schema formation and increases generalization.

5. **Episodic retrieval.** Maintain a library of previously solved programs keyed by signature features (object counts, color histograms, symmetry flags). When a new task appears, retrieve similar solved tasks and reuse or adapt their programs, akin to hippocampal retrieval of analogous episodes.

6. **Two-attempt diversity.** Inspired by the insight phase and the contest's pass@2 requirement, always produce two diverse candidate programs: one conservative (maximizing fit to the demonstrations) and one exploratory (favoring alternative plausible rules). This strategy aligns with human problem solvers who often consider multiple hypotheses.

4. Blueprint to win ARC Prize 2025

Short-term roadmap

1. **Develop a strong symbolic baseline (done).** The accompanying repository implements a fast object parser, a compact DSL and a two-step brute-force search with heuristics. It generates two predictions per test input and writes a Kaggle-ready `submission.json` without any placeholder logic. The code respects offline execution and uses only Python and NumPy.

2. **Train a neural guidance head.** Extract simple features from the training pairs (color histograms, component counts, symmetry flags) and train a small classifier (e.g., gradient-boosted trees or a tiny MLP) to predict which DSL primitives apply. Use its output to prioritize search branches. This reduces compute and improves coverage, emulating MD gating.
3. **Mine program sketches.** Mine frequent sequences of operations from the public training set and treat them as macro-operators with open parameters. Search over these sketches first and fill in parameters via beam search. This approach captures common compositional patterns.
4. **Integrate test-time adaptation (TTT).** For each task, fine-tune a tiny scoring head using the provided demonstrations and synthetic augmentations. This helps specialize heuristic scores to each task, akin to meta-learning.
5. **Implement episodic retrieval.** Maintain a database of solved programs and their signatures. At task time, compute the signature of the current task and retrieve nearest neighbors. Attempt their programs (with parameter adjustments) before doing expensive search.
6. **Continual analysis of semi-private feedback.** Use Kaggle leaderboard feedback to identify failure modes (e.g., compositional tasks, context-dependent rules) and expand the DSL sparingly. Add new primitives only when they unlock multiple unsolved tasks and avoid search explosion.

Long-term vision

- **Probabilistic program synthesis.** Replace heuristic selection with a neural generative model that samples programs proportional to their posterior probability given the demonstrations. This approach unifies search and adaptation.
- **Hybrid symbolic-neural architecture.** Combine a Vision Transformer to embed grids with a symbolic executor to run programs. The model can learn to propose program sketches directly from embeddings.
- **Broader cognitive priors.** Incorporate operations for symmetry-based filling, repetition with variable step, shape extrusion and context-dependent rules. Each addition should be grounded in an observed failure case and supported by human priors.

5. Ready-to-use solver code

A submission-ready baseline solver is packaged in the attached archive. It contains a Python package `arc_solver` and a script `arc_submit.py`. Running `python arc_submit.py` produces a `submission.json` with exactly two attempts per test input, ready for Kaggle upload. The solver uses no placeholder logic and is fully functional. See the `README.md` in the repository for details.

Conclusion

The ARC Prize 2025 challenges researchers to close the gap between human and machine reasoning. By anchoring our solver design in neuroscientific insights—MD network-driven control, basal-ganglia gating, hippocampal schema integration and oscillatory dynamics—we can build systems that adapt rapidly from few examples, prioritize the right hypotheses and produce robust solutions. The provided baseline and blueprint outline a path toward achieving the 85 % accuracy target on ARC-AGI-2 while staying within Kaggle's strict efficiency constraints.

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1 2 3 7 ARC Prize - 2025 Competition Details
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